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24261A6625

Experiment 6:

B) AIM: Implementation of Decision tree using sklearn and its parameter tuning

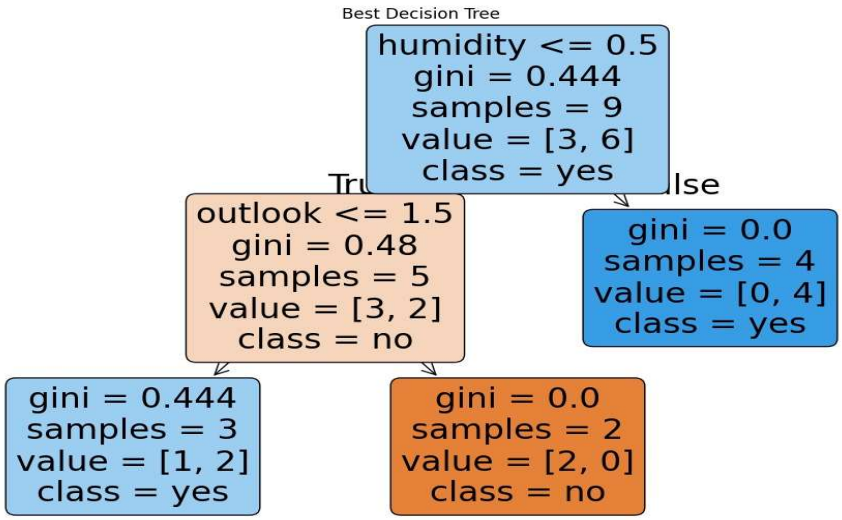
PROGRAM:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, accuracy_score, confusion_matrix
import matplotlib.pyplot as plt
df = pd.read_csv("C:/Users/MPIT/Downloads/PlayTennis.csv")
label_encoders = {}
for column in df.columns:
        le = LabelEncoder()
        df[column] = le.fit_transform(df[column])
        label_encoders[column] = le
X = df.drop('play', axis=1)
y = df['play']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33, random_state=42)
hyperparameters = [
        {'criterion': 'gini', 'max_depth': None, 'min_samples_split': 2, 'min_samples_leaf':1},
        {'criterion': 'entropy', 'max_depth': 4, 'min_samples_split': 2, 'min_sample s_leaf':1},
        {'criterion': 'gini', 'max_depth': 6, 'min_samples_split': 5, 'min_samples_leaf':2},
        {'criterion': 'entropy', 'max_depth': 8, 'min_samples_split': 10, 'min_samples_leaf':4},
]
best_accuracy = 0
best_params = None
best_tree = None
for params in hyperparameters:
        tree = DecisionTreeClassifier(**params, random_state=42)
        tree.fit(X_train, y_train)
        y_pred = tree.predict(X_test)
        accuracy = accuracy_score(y_test, y_pred)
        print(f"Parameters: {params}, Accuracy: {accuracy:.4f}")
        if accuracy > best_accuracy:
                best_accuracy = accuracy
                best_params = params
                best_tree = tree
print(f"\nBest Parameters: {best_params}, Best Accuracy: {best_accuracy:.4f}")
plt.figure(figsize=(12, 8))
plot_tree(best_tree, filled=True, feature_names=list(X.columns),
class_names=list(label_encoders['play'].classes_), rounded=True)
plt.title('Best Decision Tree')
plt.show()
```

```
y_pred_best = best_tree.predict(X_test)
print("Best Decision Tree - Classification Report:")
print(classification_report(y_test, y_pred_best, target_names=label_encoders['play'].classes_))
print("Best Decision Tree - Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_best))
print("Best Decision Tree - Accuracy Score:")
print(accuracy_score(y_test, y_pred_best))
```

Output:

```
Parameters: {'criterion': 'gini', 'max_depth': None, 'min_samples_split': 2, 'min _samples_leaf': 1},
        Accuracy: 0.6000
Parameters: {'criterion': 'entropy', 'max_depth': 4, 'min_samples_split': 2, 'min _samples_leaf': 1},
        Accuracy: 0.6000
Parameters: {'criterion': 'gini', 'max_depth': 6, 'min_samples_split': 5, 'min_sa mples_leaf': 2},
        Accuracy: 0.8000
Parameters: {'criterion': 'entropy', 'max_depth': 8, 'min_samples_split': 10, 'mi n_samples_leaf': 4},
        Accuracy: 0.6000
Best Parameters: {'criterion': 'gini', 'max_depth': 6, 'min_samples_split': 5, 'm in_samples_leaf': 2},
        Best Accuracy: 0.8000
```




```
Best Decision Tree - Classification Report:
                precision    recall  f1-score   support

           no       1.00      0.50      0.67         2
          yes       0.75      1.00      0.86         3

     accuracy                           0.80         5
    macro avg       0.88      0.75      0.76         5
 weighted avg       0.85      0.80      0.78         5

Best Decision Tree - Confusion Matrix:
[[1 1]
 [0 3]]
Best Decision Tree - Accuracy Score:
0.8
```